Sri Lanka Institute of Information Technology



Telco Customer Churn Classification and High Risk Profiling

Model Comparison, Evaluation and Implications for Retention Decisions

Final Report

IT3091 Machine Learning
Group: 2026-DS-77

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Declaration

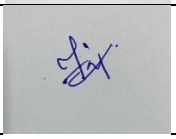
We, the members of Group 2026-DS-77, declare that this report, Telco customer churn classification and high risk profiling, describes the group investigation completed for IT3091 Machine Learning at the Sri Lanka Institute of Information Technology.

We take responsibility for the methodological decisions, submitted code and artefacts, verification of results, and interpretation presented in this report. External data, documentation and related work are acknowledged. The reported measurements come from the project evidence; analyses and outcomes that were not completed are identified as limitations or future work.

AI assisted with conceptual explanations, discussion of alternatives, code drafting and debugging, interpretation of displayed outputs, and report preparation. This assistance does not replace our responsibility to understand, check and defend the work. Details of AI use and verification are provided in Section 15.

By signing below, each member confirms that they have reviewed this revised report and accept responsibility for their stated contribution and the group submission.

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Acknowledgement

We acknowledge the IT3091 Machine Learning module team at the Sri Lanka Institute of Information Technology for providing the assignment framework and telecommunications business scenario. The Guided Data Track gave us a defined starting point while requiring us to justify our own analytical pathway.

We acknowledge the Telco Customer Churn dataset source and column documentation, the authors of the related studies cited in this report, and the developers of the open-source tools used in the implementation. Their resources supported our understanding of the data and methods; our project results were produced through our own recorded workflow.

We also acknowledge the contributions of all four group members to data understanding, preprocessing, model development, final evaluation, documentation and review. The work depended on sharing evidence and checking that each member's outputs could be used consistently by the next member.

AI assistance is disclosed in the declaration and the AI-use section. No external organisation is presented as having validated or endorsed this model.

Abstract

This project investigates telecom customer churn through the Guided Data Track, using supervised binary classification as the primary decision lens and descriptive high-risk profiling as a supporting lens. The purpose is to assess whether customer account and service information can support a useful, transparent retention shortlist.

The supplied dataset contains 7,043 records. A fixed stratified split reserved 1,409 records for final evaluation and used 5,634 for development. A shared preprocessing pipeline applied numerical parsing, training-only median imputation, categorical one-hot encoding and model-specific numerical scaling. A class-prior baseline, logistic regression, a decision tree and a random forest were compared on five shared stratified folds. Bounded tuning covered 26 configurations. The random forest had the highest mean validation average precision, 0.6662, compared with 0.6607 for logistic regression. A threshold of approximately 0.319 was selected by maximising development out-of-fold F1 and frozen before testing.

On reserved records, the selected procedure achieved average precision 0.6565, precision 0.5491, recall 0.7620 and F1 0.6383. A supplementary bootstrap analysis estimated an approximate 95% confidence interval of 0.6070–0.7087 for test AP. It flagged 519 customers, including 285 recorded churners, while producing 234 false alerts and missing 89 churners. Flagged customers tended to have shorter tenure, higher monthly charges and month-to-month contracts.

These results support a reproducible proof of concept. Unverified predictor timing, sampling and operational costs prevent claims of validated future forecasting or optimal retention action. Profiling is descriptive rather than causal. Provider-specific temporal validation and a controlled intervention study are recommended before deployment.

Keywords: customer churn; supervised classification; average precision; random forest; cross-validation; high-risk profiling.

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List of Abbreviations

Abbreviation	Meaning
AI	Artificial intelligence
AP	Average precision
CI	Confidence interval
CLV	Customer lifetime value
CSV	Comma-separated values
CV	Cross-validation
DSL	Digital subscriber line
EDA	Exploratory data analysis
F1	Harmonic mean of precision and recall
FN	False negative
FP	False positive
ID	Identifier
JSON	JavaScript Object Notation
KNN	K-nearest neighbours
ML	Machine learning
OOF	Out-of-fold
PR	Precision–recall
ROC-AUC	Area under the receiver operating characteristic curve
SD	Standard deviation
SHA-256	Secure Hash Algorithm with a 256-bit digest
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Oversampling Technique
SMOTENC	SMOTE for nominal and continuous features
SVM	Support vector machine
TN	True negative
TP	True positive

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1. Introduction

Customer churn is the loss of customers who stop using a provider's services. For a telecom retention team, useful analysis must do more than produce a high overall score: it should identify a manageable set of customers for attention, show the errors in that identification and explain what the resulting group looks like. This project addresses that decision through the supplied Telco Customer Churn dataset and a reproducible machine-learning workflow.

We compare a simple reference with three supervised classifiers, select a procedure using development data, and evaluate it on reserved records. A secondary profiling analysis describes the customers flagged by that same procedure. The report explains the evidence and trade-offs behind the choices, rather than treating the workflow as a fixed recipe.

1.1 Business Context

The assignment scenario describes a telecom provider seeking to reduce customer loss and improve retention decisions [1]. The intended stakeholder is its customer retention or marketing team. Customer records include contract, service, payment and charge information, which may help distinguish patterns associated with recorded churn. Related Sri Lankan studies illustrate the relevance of service and tariff questions, but do not establish that this supplied dataset represents a Sri Lankan provider [5–8].

The scenario supplies no verified outreach budget, contact capacity, intervention success rate or customer profitability. We therefore formulate a decision-support study rather than a profit-maximisation system. A provider would need those operational details before turning a model shortlist into an outreach policy.

1.2 Business Problem

Treating every customer as equally likely to leave can spread retention attention too widely. Conversely, a very restrictive shortlist can miss customers who would have benefited from attention. The analytical problem is to distinguish customer records associated with churn while making this trade-off visible through precision, recall, false alerts, missed churners and flagged workload.

The available label records whether a customer left within the last month; the data do not establish that every input preceded that outcome. Our exact task is therefore classification of recorded churn on held-out customers. It provides an initial proof of concept for possible future retention support.

1.3 Project Aim and Objectives

The aim is to develop and critically evaluate a reproducible customer-level churn classification procedure, then use its flags to provide descriptive evidence for a cautious retention recommendation.

- Verify the source, variable meanings and data quality, and preserve a fixed development/test split so later comparisons use consistent evidence.
- Investigate development patterns and translate them into justified preprocessing choices without using test outcomes to guide development.
- Compare a class-prior baseline and three classifier families through complete pipelines on common validation folds.
- Select the model by mean validation AP and an operating threshold by development OOF F1, then freeze both before test evaluation.
- Measure ranking, classification errors and workload on reserved records and describe the flagged and unflagged groups.
- Document decisions, limitations, reproducibility, AI assistance and individual contributions, and recommend what further evidence is needed for operational use.

1.4 Scope of the Completed Solution

The completed solution comprises four core executed notebooks and a supplementary uncertainty-analysis notebook, a reusable preprocessing module, saved data splits and cross-validation membership, an evaluated fitted random-forest pipeline, a frozen threshold, exported results and figures, and shared decision and technical logs. It uses 19 original predictors and produces a churn score, a high-risk flag and descriptive group profiles. The baseline and three alternatives were compared; 26 configurations were tuned before final testing.

The work does not include a deployed customer-facing application, a validated future prediction horizon, causal explanations, individual feature-attribution analysis, customer lifetime value, an optimised retention offer, or measured retention gains. Separate clustering, resampling and preprocessing ablation experiments were not completed. These boundaries keep the claim aligned with the evidence and identify meaningful future work.

2. Problem Framing and Decision Lenses

Table 1: Problem framing canvas

Canvas element	Our formulation
Stakeholder	Telecom provider customer retention or marketing team.
Decision need	Identify higher-scoring customer records, understand the flagged group and assess the errors and workload of a possible retention shortlist.
Unit of analysis	One individual customer per record. Customer-level outreach is the intended decision context.
Primary lens	Supervised binary classification of recorded Churn
Secondary lens	Descriptive high-risk customer profiling using the same model scores and frozen threshold.
Exact outputs	A score for Churn = Yes, a threshold flag, measured held-out performance and flagged-versus-unflagged profiles.
Selection criterion	Highest unrounded mean development cross-validation average precision.
Operating rule	Score ≥ 0.319015704046037, selected by maximum development OOF F1.
Evaluation boundary	Held-out records from this dataset; no verified future-period, Sri Lankan provider or intervention-impact claim.

2.1 Why these lenses fit together

The primary lens directly uses the available binary target and creates a measurable ranking and classification task. Profiling alone could describe differences without establishing how well a model identifies recorded churners. A separate segmentation objective would require its own grouping rationale and validation, and would not directly answer our chosen classification question.

The secondary lens is useful because a retention team needs to understand the shortlist as well as its error rate. It reuses the primary model output, rather than introducing a disconnected analysis. It is optional in principle; we included it because the added descriptive evidence has a clear role. The profiles describe group composition, not individual prediction explanations or proven reasons for churn.

2.2 Task statement

We use customer account and service information to classify recorded churn and produce a model score and frozen-threshold flag for each evaluated customer. We assess the resulting errors and describe the flagged group to support a cautious, evidence-based retention recommendation.

Evidence: Group Decision Log D01–D06 and D04-R1; assignment descriptor [1].

3. Scope choices and related work

3.1 Why the Guided Data Track

Guided Data allowed us to investigate an explicit decision problem while choosing our own lenses, preprocessing, models and evaluation. Industry Explorer requires prior approval, a suitable real-world problem and evidence of engagement and data handling. Those additional conditions were not established for this telecom investigation; finding relevant papers or a public dataset would not satisfy them [1].

3.2 How the proposal changed

An earlier draft considered combining churn scores with MonthlyCharges × tenure as a customer-value proxy. This idea was removed before the initial submission dated 16 September 2026, which explicitly excluded CLV-driven prioritisation and combined risk-value ranking. Current monthly charges multiplied by tenure cannot establish future customer value or profit because historical charges, service costs and future customer duration are not established. The completed project retained this exclusion, recorded in D04-R1.

We did not adopt a fixed top-10-percent contact rule. Our final high-risk definition uses the development-selected threshold. No verified budget, contact capacity or financial cost matrix was available. We also avoided an independent clustering pipeline: existing contract and service categories were sufficient for the selected descriptive profiling lens.

3.3 What the background reading contributes

Adikari, Rabel and Samarasinghe studied Sri Lankan mobile churn through a questionnaire, with 305 responses from 400 distributed questionnaires [5]. This provides local business context and illustrates that tariffs and service perceptions merit stakeholder investigation. It does not validate the population, labels or predictive performance of our supplied dataset.

Alwis, Kumara and Hapuarachchi considered several analytical approaches and classifiers for telecommunications decision-making [6]. This supports considering alternative methods, but their data and evaluation are different. We did not transfer their performance claims to our experiment or copy their broader combination of clustering and association analysis.

The available two-page 2025 dissertation extract by Abayakoon contains a title page and abstract [7]. It describes classification methods and interpretability in Sri Lankan fixed telecommunications. We treated it as contextual reading, not a verified performance benchmark: the full methodology and validation detail were not available in that extract. The Gamage repository abstract [8] was likewise background reading, not a basis for selecting our model.

Our contribution is a traceable, coherent comparison on the supplied data. No claim is made that this is a new algorithm, a globally best churn model, or an operationally validated Sri Lankan system.

4. Workflow and responsibility handovers

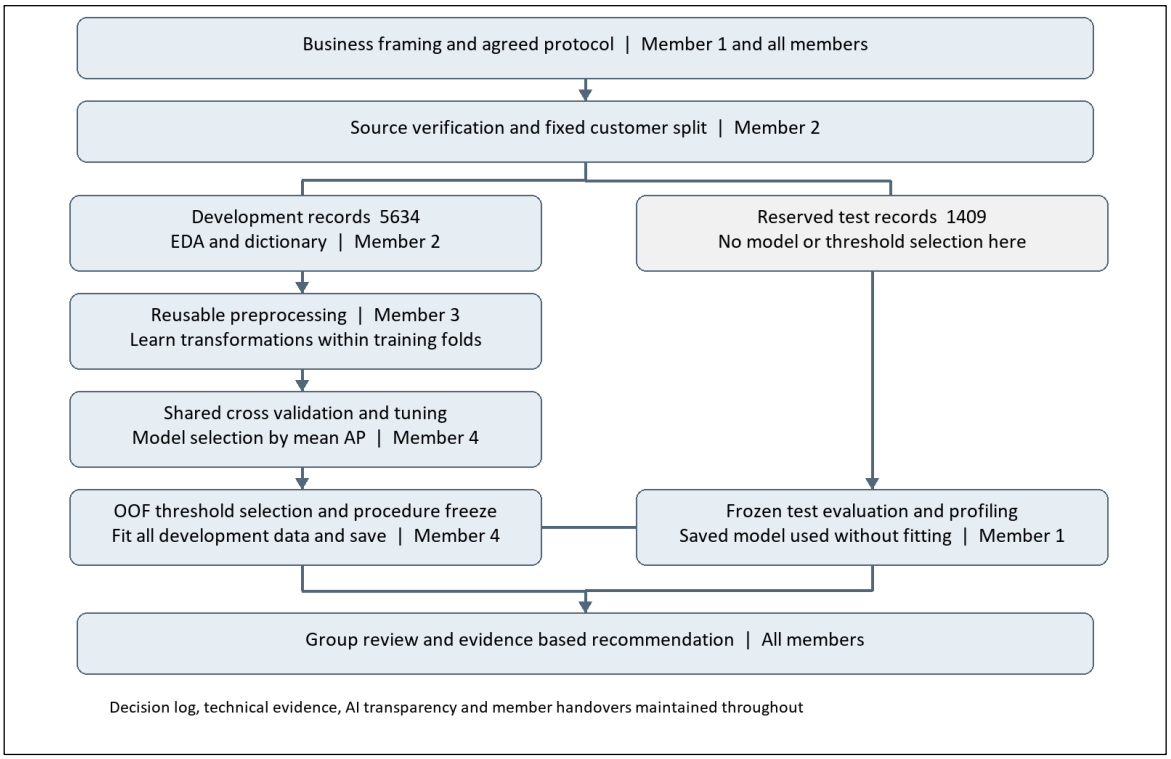


Figure 1: Implemented workflow. The reserved test branch joins only after the procedure is frozen.

The workflow follows dependencies while allowing discussion and documentation throughout. Development EDA informs preprocessing choices. Member 3 supplies reusable, unfitted definitions; Member 4 fits them inside each training fold. After model and threshold selection, a fresh pipeline is fitted on all development records and handed to the leader for final evaluation.

The leader checks the handover files, scores the reserved test without fitting and joins predictions to original records by customerID for profiling. Group review then connects the statistical results, practical workload and limitations to the final recommendation. Test results were not used to revise the model or threshold. A supplementary analysis subsequently estimated bootstrap confidence intervals from the saved test predictions without changing the fitted model, threshold or predictions.

Table 2: Member handover evidence

Handover	Required evidence
Member 2 to Member 3	Original fingerprint; saved customer split; development data; dictionary; EDA findings and unresolved metadata.
Member 3 to Member 4	Shared parser and preprocessor factory; diagnostic checks; feature policy; environment and file fingerprints.
Member 4 to Member 1	Fitted pipeline; frozen settings and threshold; selection and OOF results; reload check; handover record.
Leader to group	Final metrics and predictions; bootstrap uncertainty results; profiles; output inventory; limitations and stakeholder recommendation.

Evidence: Core notebooks 01–04 and supplementary notebook 05; shared module; three logs; bootstrap uncertainty results; model and evaluation records.

5. Dataset Provenance and Quality

We used the supplied file WA_Fn-UseC_-Telco-Customer-Churn.csv, associated with the Telco Customer Churn dataset published by BlastChar on Kaggle [2]. Each row represents one customer. The file contains 7,043 records and 21 columns: one customer identifier, one target variable and 19 candidate predictors.

The original CSV was preserved unchanged. Its SHA-256 fingerprint was recorded so that subsequent stages could verify the identity of the source file. Saved development/test membership was reused to maintain a consistent evaluation split. The data dictionary distinguishes definitions supported by the reviewed documentation from characteristics observed in our development data. The complete data dictionary is submitted separately as outputs/tables/development_data_dictionary.csv. It records documented variable meanings, observed development values, missingness, source limitations and final handling notes consistent with the implemented preprocessing pipeline.

Table 3: Dataset quality checks

Check	Verified result and implication
Dimensions	7,043 rows, 21 columns: one identifier, one target and 19 predictors.
Identifiers and duplicates	No missing customer IDs, repeated customer IDs or exact duplicate rows. No duplicate-removal step was required.
Target	No missing target values; categories Yes and No. Modelling uses Yes = 1 and No = 0.
TotalCharges parsing	11 blank/missing amounts in the source; no unexpected non-numeric text. Eight blanks are in development and three in test.
Development categorical quality	No missing values or blank strings in the 16 categorical predictors.
Service consistency	All seven tested service-state relationships had zero inconsistent development records.
Dictionary review	Column definitions were checked against the supplied Kaggle metadata on 23 September 2026. Details not established by that documentation are recorded as unresolved.

5.1 Construction and temporal limitations

The source description defines Churn as whether a customer left within the last month. However, the supplied file and reviewed documentation do not establish the exact calendar period, the detailed sampling procedure for churners and non-churners, or whether every predictor was measured before the churn outcome period. A maximum tenure of 72 months describes customer relationship duration; it does not establish six years of dated observations.

The observed churn proportion is a sample characteristic, not a verified provider monthly churn rate. The reviewed documentation also does not establish Sri Lankan operational representativeness, the currency of the charge fields or the exact age boundary used for SeniorCitizen. We therefore report charges as recorded amounts and do not interpret Partner as confirmed marital status.

These limitations restrict the claims supported by our evaluation. The reserved test set assesses classification of recorded churn on held-out customers from the supplied dataset. It does not establish future-period forecasting performance. Validating that use would require dated customer histories, features available before a defined prediction date, future outcome labels and evaluation on a later period.

5.2 Source fingerprint

A SHA-256 fingerprint identifies the contents of a file. We recorded the original CSV's fingerprint and verified that it matched before final evaluation. This check supports reproducibility by confirming that the source file remained unchanged; it does not establish the accuracy or representativeness of the data.

Recorded SHA-256:

88be4b93fbe0cc83421af1c503794c97c342eca914c1576db7c276e61d61358a

File-verification and split evidence: Notebook 01 and split_record.json.

Column-definition and documentation-review evidence: development_data_dictionary.csv and dictionary_source_review.csv.

6. Development Data Exploration

6.1 Fixed split and class distribution

Table 4: Development and test class distribution

Subset	Customers	Churn Yes	Churn No
Development	5,634	1,495 (26.54%)	4,139 (73.46%)
Reserved test	1,409	374 (26.54%)	1,035 (73.46%)

The dataset was divided into 5,634 development records and 1,409 reserved test records using stratification by Churn, test_size = 0.2 and random seed 42. Stratification preserved approximately the same churn proportion in both subsets.

Customer membership was saved in data/splits/split_manifest.csv, relative to the shared project folder, ML 2026-DS-77 Telco Customer Churn. This file records each customerID and its assigned subset (development or test). The saved membership was reused throughout the project to maintain the same split. Detailed exploratory analysis after splitting used a separate copy of the development data. The test counts in Table 4 are included for final reporting; they were not used to select preprocessing or model settings.

Before adopting the split protocol, the group had viewed whole-file descriptive summaries. This represents prior human exposure to aggregate information, not model training on test records. The reserved test records were subsequently excluded from fitting preprocessing, model tuning and threshold selection. We therefore acknowledge that test information was not completely unseen by the group, while maintaining its reserved role in final model evaluation.

The 80/20 allocation retained most records for development while providing 1,409 customers for a separate final assessment. This was a practical choice rather than an experimentally established optimal ratio. The fixed seed supports reproducibility; it was not selected to improve performance. A chronological split would better assess future-period performance, but verified observation dates were unavailable. Neither customerID nor tenure establishes calendar order.

Always predicting “No” would achieve 73.46% development accuracy while identifying no churners. This motivated average precision for model selection and precision, recall and F1 for evaluating classification decisions, rather than relying on accuracy alone. The unequal class proportions did not, by themselves, establish a need for resampling.

6.2 Numerical distributions

We examined numerical distributions to identify missingness, unusual values and differences in feature scale, and to inform preprocessing decisions. Table 5 summarises observed development values before imputation. “Observed n” denotes the number of non-missing values; medians and ranges exclude missing values. Charge values are reported as recorded amounts because their currency was not established in the reviewed documentation.

Table 5: Development numerical summary

Feature	Observed n	Median	Range	Main observation
tenure	5,634	29	0–72	Concentrations at short and long durations
MonthlyCharges	5,634	70.50	18.40–118.75	Several concentrations
TotalCharges	5,626	1,398.125	18.85–8,684.80	Right-skewed; eight missing

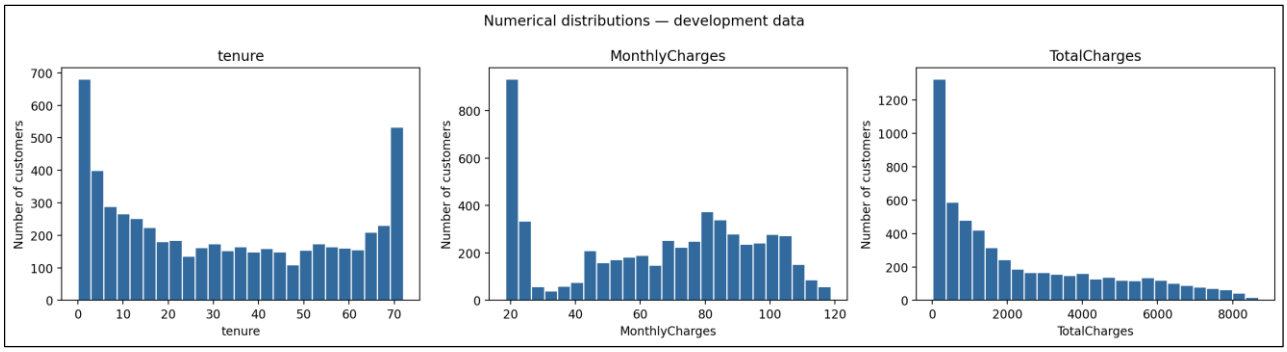


Figure 2: Numerical distributions in development data only.

All eight missing development TotalCharges values occurred among customers with tenure = 0. This identifies a missingness pattern but does not establish that their true total charge was zero. We retained these customers to avoid removing all records with missing totals, which represented a specific tenure group. Missing amounts were subsequently handled through median imputation fitted only on the relevant training data, as described in Section 7.

The distribution of TotalCharges was right-skewed, with many smaller amounts and a tail of larger values. Skewness and points beyond boxplot whiskers do not, by themselves, establish data errors. We therefore did not automatically remove or cap these observations. Alternative outlier treatments were not separately compared.

Evidence: EDA-01, EDA-02 and EDA-05; development_numeric_summary.csv; Notebook 02 for the implemented imputation procedure.

6.3 Development associations and interpretation

Table 6: Contract and recorded churn

Contract	Customers	Recorded churners	Churn percentage
Month-to-month	3,102	1,326	42.75%
One year	1,173	130	11.08%
Two year	1,359	39	2.87%

Churn percentages are calculated within each contract category. For example, 1,326 of 3,102 month-to-month customers had recorded churn, giving 42.75%. These are descriptive sample associations, not estimates of the causal effect of contract type.

Contract term showed a substantial unadjusted association with recorded churn. Churners also had shorter median tenure (10 versus 38 months), higher median monthly charges (79.95 versus 64.40) and lower median observed total charges (740.30 versus 1,691.90). The distributions overlap; these are not deterministic classification rules.

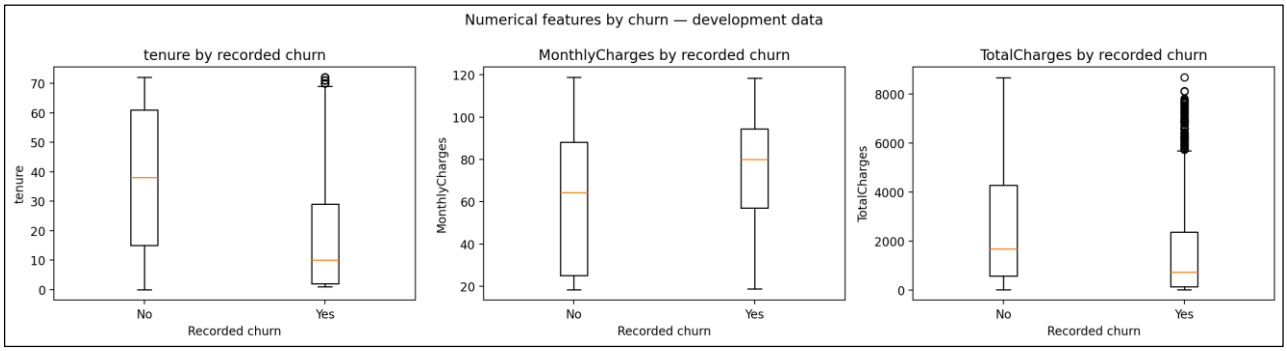


Figure 3: Development numerical distributions grouped by recorded churn.

Electronic-check customers had 45.74% recorded churn, compared with 14.92–19.28% for other payment methods. Fiber-optic customers had 42.09%, compared with 18.69% for DSL and 7.25% without internet service. Within each contract category, electronic-check churn proportions remained higher, but other characteristics were not controlled. The two-year electronic-check group contained only 10 recorded churners, so comparisons involving this subgroup should be interpreted cautiously.

Repeated “No internet service” categories describe the same 1,214 development customers, including 88 churners. These are overlapping descriptions, not independent replications. Gender churn proportions were similar, at 26.63% and 26.44%; that observation alone neither establishes fairness nor proves irrelevance to a multivariable model.

Table 7: Correlations between numerical predictors

Pair	Pearson	Spearman	Records
tenure and TotalCharges	0.829	0.891	5,626
MonthlyCharges and TotalCharges	0.654	0.642	5,626
tenure and MonthlyCharges	0.257	0.284	5,634

The correlations show overlapping information. They do not establish an exact billing formula or justify automatic feature deletion. No formal hypothesis tests or causal effects are claimed from this EDA.

We used both Pearson and Spearman summaries because they answer related but different questions: Pearson describes linear association, while Spearman describes monotonic association using ranks. Reporting pair counts makes the effect of missing TotalCharges visible. Neither measure evaluates whether one feature adds information conditional on all others, so high pairwise correlation was not used as an automatic removal rule.

Evidence: EDA-06–EDA-10 in the EDA Insight Log;
development_contract_churn.csv,
development_numeric_by_churn.csv,
development_categorical_churn.csv,
development_payment_by_contract.csv,
development_pearson_correlations.csv,
development_spearman_correlations.csv and development_correlation_pair_counts.csv.

File locations are listed in the evidence inventory.

7. Preprocessing and Feature Decisions

We implemented one shared preparation function for the 19 original predictors. customerID is retained separately for traceability and joins; Churn is used only as the target. SeniorCitizen is treated as categorical despite integer storage. Three numerical and 16 categorical predictors produce 46 encoded features in the fitted development pipeline.

7.1 Numerical parsing and missing values

The parser strips TotalCharges whitespace, converts valid amounts to numeric and preserves documented blanks as missing values. Unexpected non-numeric content causes a failure instead of silent acceptance. A SimpleImputer learns the median of each numerical feature from the training records available to that fit. Validation and test records use those learned values.

We retained the eight affected development records because all had tenure zero; deleting them would remove customers from a specific observed group. Filling the amounts with zero would assume that a blank means no accrued charges, which the reviewed documentation did not establish. We selected median imputation as a simple, reproducible treatment that is less sensitive to large charge values than mean imputation. Because missing totals affected only 8 of 5,634 development records (approximately 0.142%), we kept this treatment fixed while prioritising model and class-weight comparisons. This limited the experimental scope but did not establish median imputation as optimal. In particular, the median may be unrealistic for new customers. Alternative replacements and an additional missingness indicator were not tested; a targeted sensitivity comparison could strengthen this decision.

7.2 Categories and scaling

OneHotEncoder used drop=None, handle_unknown="ignore" and dense output. This avoids arbitrary numerical ordering and preserves “No internet service” and “No phone service” as meaningful states. No categorical missing values or blank strings were observed in development data, so we did not introduce categorical imputation. The parser rejects such inputs if encountered, making unexpected data visible rather than silently assigning a replacement. Operational use would require an agreed policy for handling these rejected records.

Arbitrary integer codes would impose a numerical order on nominal categories such as payment method. Collapsing “No internet service” into “No” would erase the distinction between not having internet and having internet without an add-on. We retained all indicator columns for one consistent representation across the regularised logistic and tree-based pipelines. This can create redundant columns; dropping a reference category is a reasonable alternative, but it was not separately tested. Dense output was manageable for 46 features and simplified verification; it was not selected through a speed benchmark.

Logistic regression used StandardScaler on numerical variables after imputation. The scaler learned means and standard deviations from the training records available to each fit. This made numerical scales comparable so that coefficient regularisation was not driven by differences in measurement units. The dummy, decision tree and random forest pipelines retained unscaled numerical values; standardisation was unnecessary for the tree models' ordering-based splits. We kept standard scaling fixed to focus the comparison on model settings and class weighting. Min–max and robust scaling were not tested, so the results compare the implemented pipelines rather than every possible preparation of each algorithm. Standardisation does not remove skewness or make unusual observations harmless.

7.3 Feature retention and imbalance

We retained all 19 original predictors as a manageable starting representation. EDA identified overlapping information but did not establish that any predictor should be removed; the predictive effect of feature removal would require a separate validation comparison. Correlated charges and tenure can still contain different information; automatic deletion could remove useful variation. Similarly, extreme but plausible charges may describe real customers, so deleting or clipping them solely because a boxplot marks them would change the population being modelled. No distributional assumption required us to make TotalCharges normally distributed. These arguments support a conservative starting policy; without separate ablations, they do not establish that feature removal, transformations or engineered predictors could not help.

No additional business-derived predictors were included in the final pipeline. The initially proposed MonthlyCharges × tenure value proxy was not used. One-hot encoding changed the representation of the existing predictors; it did not introduce a customer lifetime value estimate.

We treated imbalance as an empirical modelling question rather than a requirement to force a 50/50 class split. Development data contained 1,495 churners, and each CV training fold retained 1,196 churners. This gave a meaningful starting sample for unweighted learning, although the count alone cannot prove that imbalance is harmless. The following section explains why class weighting was compared while resampling was left outside the bounded experiment.

Evidence: PP01–PP06; D12–D14; src/telco_preprocessing.py.

7.4 Why class weighting was compared without resampling

The available classes were unequal, but the business objective was to identify recorded churners with acceptable errors, not to make the training class counts equal. We evaluated that objective with AP and threshold-specific precision and recall. Starting with the original training records let us measure what the classifiers could learn before adding a sampling procedure.

Random undersampling would discard observed non-churners. For this moderate-sized dataset, retaining those examples preserved information about customers the model should not flag. Random oversampling would retain all observations but duplicate some minority records; this changes their influence without adding independent customer evidence. Either approach could still help, but both would add a sampling ratio and another experimental choice.

Ordinary SMOTE generates interpolated examples. With 16 categorical predictors, applying it directly to one-hot columns can create fractional category indicators that do not describe an observed category assignment. A mixed-data method such as SMOTENC is a possible alternative [9], but it would need a suitable pipeline order, neighbour settings and checks on service consistency. We did not run that experiment. These additional design requirements placed synthetic resampling outside the planned comparison; they do not establish that categorical-aware oversampling would be ineffective.

Class weighting offered a smaller, direct comparison: retain the same rows but alter the influence of the two classes during fitting. The balanced option uses weights inversely related to class frequencies in the current training data [10]. For each of the three classifier families, the search included balanced and unweighted versions of each combination of the other searched settings, evaluated on the same validation folds. This helped isolate the weighting choice without simultaneously changing the training sample.

Within the evaluated grid, the best configuration in each family was unweighted. That supports the selected unweighted procedure under the agreed mean-AP criterion. It does not show that weighting cannot improve recall at another threshold, that resampling would perform worse, or that the observed sample prevalence equals a provider's prevalence.

The development OOF threshold was selected separately because changing the operating threshold adjusts the precision–recall trade-off without synthesising or deleting training customers. Threshold selection is not a universal substitute for class weighting or resampling; it changes the decision rule rather than the learned score ranking. Any future resampling comparison must occur within training folds, preserve validation/test distributions and use a new evaluation plan now that this test set has been examined.

7.5 Preprocessing verification and leakage controls

7.5.1 Development diagnostic checks

Table 8: Preprocessing verification results

Check	Result	Meaning
Diagnostic split	4,507 training; 1,127 validation	Uses development records only.
Scaled and unscaled output	46 features for both variants	Rows preserved; no missing or infinite transformed outputs.
Training medians	29; 70.35; 1,377.975	tenure, MonthlyCharges and TotalCharges respectively; learned from diagnostic training only.
Numerical centring	Passed for scaled variant	Checks scaling behaviour on diagnostic training data.
Unseen PaymentMethod	Transformation succeeded; indicators all zero	Checks the specified unknown-category fallback.
Shared module equivalence	Both variants matched prior diagnostic outputs	Reusable code agrees with notebook implementation.
Input preservation	Original predictors unchanged	Parsing and diagnostics operate on copies.

The medians in Table 8 were learned from the 4,507 diagnostic training records. They are diagnostic results, not fixed replacement values imposed on every subsequent cross-validation fold.

The artificial unseen-category check verified that transformation succeeded under the configured fallback. An unfamiliar category produces zeros across that feature's indicator columns, but the model has not learned a specific representation for that category. Successful transformation therefore does not establish reliable prediction. Unfamiliar categories would require monitoring in operational use.

7.5.2 Why the shared module matters

make_model_preprocessor returns a new, unfitted pipeline. Its fixed parser learns no statistics; its imputers, scaler and category encoder learn during each fit. Member 4 placed this preparation inside the full classifier pipeline and cloned the complete pipeline for each fold. This prevents validation records from influencing learned medians, scales or category mappings [3]. It does not resolve the source dataset's unknown timing of predictors relative to churn.

We handed over reusable code rather than a preprocessor fitted on all development records. Fitting preparation before cross-validation would allow validation information into the training process. The final development fit legitimately learns from all 5,634 development records only after selection. The three missing test TotalCharges values were later filled internally using that development-learned median.

7.5.3 Evidence retained

The preprocessing evidence records missingness, diagnostic transformations, unfamiliar-category handling and agreement between the notebook implementation and shared module. A fresh-session, top-to-bottom execution completed successfully. These checks verified that the chosen preparation behaved as intended and could be handed over reproducibly. Comparing alternative preprocessing policies would require additional validation experiments; the checks do not establish that the selected policy was optimal.

Evidence: Notebook 02; src/telco_preprocessing.py; preprocessing_missingness.csv; preprocessing_diagnostic_checks.csv; preprocessing_unknown_category_checks.csv; preprocessing_module_checks.csv; and preprocessing_run_record.json.
File locations are listed in the evidence inventory.

8. Validation Protocol and Evaluation Measures

8.1 Shared development folds

We used five stratified folds with shuffling and seed 42. Each development customer appears in validation exactly once. Four folds contain 4,507 training and 1,127 validation records; the fifth contains 4,508 and 1,126. Each validation fold has 299 churners. The saved manifest and settings ensure every candidate is evaluated on the same customers.

Five folds balance repeated evaluation with a manageable search cost: each fit uses about 80% of development records, and every record supplies one validation result. A single validation split would make the comparison depend more on one partition, while ten folds would approximately double the number of tuning fits. We did not test fold counts as hyperparameters; the shared five-fold design was fixed for consistent comparison.

A complete fresh pipeline is fitted on each fold's training portion and scored on that fold's validation portion. Training AP was also recorded to identify training–validation gaps that may indicate overfitting. These gaps were interpreted alongside validation performance rather than treated as proof on their own. Fold sharing makes comparisons consistent, but the folds are not independent repeated studies, and their standard deviation is not a confidence interval.

8.2 Why average precision was primary

Average precision (AP) summarises precision across increases in recall as the score threshold varies. It assesses positive-class ranking without requiring one operating threshold. scikit-learn computes a recall-weighted precision sum; it is not simply trapezoidal area under the precision–recall curve [4].

Mean validation AP was the arithmetic mean of the five fold AP values, and unrounded values determined selection. AP was suitable for prioritising churn identification when a majority-class accuracy score would hide failure to identify any churners. AP is not accuracy, calibration or financial return. Its interpretation also depends on the class distribution.

Accuracy can conceal missed churners because non-churners are the majority. ROC-AUC can also describe ranking, but it summarises true-positive versus false-positive rates rather than the precision of a positive shortlist. AP more directly matched our interest in identifying churners among flagged records. We did not reject ROC-AUC as invalid; we selected one primary criterion in advance to avoid switching metrics to favour a result. F1 was reserved for threshold selection because it depends on an operating cut-off.

Throughout the evaluation, Churn = Yes is the positive class. TP denotes correctly flagged churners, FP denotes flagged non-churners, FN denotes missed churners and TN denotes correctly unflagged non-churners.

Table 9: Evaluation measures and purposes

Measure	Definition and purpose
Average Precision (AP)	Summarises precision weighted by increases in recall across score thresholds. The arithmetic mean of five validation-fold AP values was the primary model-selection criterion.
Precision	TP / (TP + FP): among flagged customers, how many have recorded churn?
Recall	TP / (TP + FN): among recorded churners, how many were flagged?
F1	2TP / (2TP + FP + FN): harmonic balance of precision and recall at a threshold.
Confusion matrix	Separates correct detections, missed churners, false alerts and correct non-churn classifications.
Flagged count and proportion	Measures the size of the model-generated shortlist and the implied contact workload.
Precision–recall curve	Shows threshold trade-offs; the frozen threshold is marked for the final procedure.

A default 0.50 cut-off has no established business meaning here, particularly without calibration and verified error costs. A fixed top percentage would assume an outreach capacity that the scenario does not provide. F1 supplies an explicit compromise between precision and recall, but it embeds its own preference and ignores true negatives. The chosen threshold is therefore a transparent academic operating rule, not a claim about equal monetary costs or an optimal contact budget.

Precision alone could favour a very small shortlist that misses many churners, whereas recall alone can be maximised by flagging everyone. An F-beta measure could place greater emphasis on precision or recall, but the scenario did not establish which preference or weight to use. We therefore adopted F1 as an explicit development criterion giving equal emphasis to precision and recall in its formula. This was a chosen compromise, not a neutral or financially optimal policy.

Evidence: D09–D11; model_development_protocol.json; development_cv_manifest.csv; development_cv_settings.json. Locations are listed in Appendix E.

9. Candidate Models and Bounded Search

Table 10: Candidate model rationale

Candidate	Why included	Important constraint
Dummy prior baseline	Constant class-prior score measures performance without individual discrimination.	AP is close to positive-class prevalence.
Logistic regression	Regularised additive comparator with a relatively simple structure.	May miss nonlinear relationships and interactions unless represented.
Decision tree	Can learn nonlinear branching rules and interactions.	Single-tree estimates can vary and overfit; depth and leaf sizes constrain complexity.
Random forest	Ensemble of trees can capture interactions with reduced single-tree variability.	Less directly interpretable; complexity must earn its performance gain.

The selected set provides distinct comparisons: a no-discrimination reference, a regularised additive model, a single nonlinear tree and a tree ensemble. This makes the performance-versus-complexity question observable while satisfying the baseline-plus-three-alternatives requirement. We did not select families only because they are popular or because another paper reported high accuracy.

KNN would introduce a distance-based approach requiring additional choices about similarity across numerical and categorical attributes. SVM would introduce kernel and regularisation choices; boosting would introduce a different sequential ensemble and its tuning requirements; and a neural network would require architecture and optimisation decisions. We prioritised examining the selected families and their complexity and class-weight settings rather than extending the number of families. This limits the conclusion to the evaluated pipelines. In particular, boosting remains a credible alternative that could perform better; it was not excluded on the basis of measured inferiority.

9.1 Initial settings and tuning

Initial logistic regression used L2 regularisation, C = 1, lbfgs and max_iter = 2000. The initial tree used depth 4 and minimum leaf size 20. The initial forest used 200 trees, unrestricted depth, minimum leaf size 5 and sqrt feature sampling. Tree-based models used seed 42, and the forest used n_jobs = 1. Initial class weighting was None.

Table 11: Bounded tuning search

Family	Ordered settings varied	Configurations	CV fits
Logistic regression	C: 0.1, 1, 10; weight: None, balanced	6	30
Decision tree	Depth: 3, 4, 6; leaf: 20, 10; weight: None, balanced	12	60
Random forest	Depth: 10, None; leaf: 10, 5; weight: None, balanced	8	40

For logistic regression, C values of 0.1, 1 and 10 tested stronger, reference and weaker regularisation on a logarithmic scale. For the single tree, depths of 3, 4 and 6 tested increasingly detailed branching while leaf minima of 10 and 20 restricted very small terminal groups. For Random Forest, depth 10 versus unrestricted depth and leaf minima of 5 versus 10 compared more and less constrained trees. These were deliberately limited contrasts, not a search over all plausible values.

The saved protocol specifies the ordered configurations before comparison. Exact within-family ties select the first saved configuration. Exact between-family ties follow dummy, logistic regression, decision tree, then random forest. The search was not expanded in response to the final test results. Appendices A-C report all 26 configurations.

The bounded grid examined logistic regularisation, tree depth, minimum leaf size and class weighting. Enumerating it ensured that every listed configuration used the same folds. Training and validation AP were both examined for possible overfitting. Other settings remained fixed to keep the comparison interpretable. The forest used 200 trees as a practical fixed choice; no tree-count convergence study was performed. The results therefore support selection within the stated grid rather than exhaustive optimisation.

The initial comparison used 20 fits, tuning used 130, selected-configuration OOF scoring used five and the final development fit used one: 156 classifier fits per complete modelling run. Diagnostics and later reload checks are separate from this count.

9.2 Development comparison results

Table 12: Initial cross validation comparison

Initial model	Training AP	Validation AP	Fold SD	AP gap
Logistic regression	0.6675	0.6607	0.0194	0.0068
Random forest	0.8343	0.6592	0.0268	0.1751
Shallow decision tree	0.6146	0.6018	0.0250	0.0128
Dummy prior	0.2654	0.2654	0.0001	0.0000

The initial forest showed a large training–validation gap of about 0.1751. Its training AP of 0.8343 did not translate into superior validation ranking relative to logistic regression. This is why selection used validation evidence rather than the training score.

Table 13: Best searched model comparison

Best searched model	Mean train AP	Mean valid AP	Fold SD	AP gap
Random forest	0.7568	0.6662	0.0244	0.0905
Logistic regression	0.6675	0.6607	0.0194	0.0068
Shallow decision tree	0.6836	0.6124	0.0213	0.0712

9.3 Selected configuration

Random forest configuration 1 had the highest unrounded mean validation AP, 0.666248. It used 200 trees, maximum depth 10, minimum samples per leaf 10, sqrt feature sampling and no class weighting.

Other settings include bootstrap=True, Gini criterion, minimum samples split 2, random_state=42 and n_jobs=1. The complete settings are preserved in frozen_model_procedure.json.

The best logistic regression used C = 1 with no class weighting (AP 0.660713). The best tree used depth 6 and minimum leaf size 20, also unweighted (AP 0.612360). Restricting the forest reduced its training–validation gap to 0.0905 while improving validation AP relative to the initial forest.

Depth and minimum leaf size changed together between the initial and selected forest configurations. Their individual contributions cannot be isolated from this before-and-after comparison alone.

9.4 How strong is the preference for the forest

The forest exceeded logistic regression by approximately 0.0055 AP. We followed the recorded selection rule, but this small advantage does not prove a meaningful business gain or statistical superiority. Logistic regression remains an important simpler comparator. Neither overlapping fold variation nor the mean difference alone establishes a significance result; no formal paired significance test was performed.

Changing the selection rule after observing this close result would weaken the traceability of the experiment. We therefore retained the forest selected by the agreed AP rule and reported the practical counterargument: logistic regression was simpler, faster in this run and almost as strong on development AP. If a stakeholder later prioritises interpretability or resource use, that should become an explicit criterion in a new evaluation, rather than an unsupported claim that the forest is universally preferable.

Mean fit times in the saved tuning run were about 0.10 seconds for the best logistic model, 0.09 for the tree and 1.27 for the forest. These environment-specific timings illustrate a complexity trade-off, not a production latency benchmark. Search reuse of the same folds also means these are development selection estimates, not independent final performance estimates.

Evidence: initial_model_cv_summary.csv; tuned_model_cv_best_per_model.csv; selected_model_configuration.json.

10. Threshold selection and model freezing

We regenerated one out-of-fold score for each of the 5,634 development customers using the selected forest configuration and the same five folds. Each score came from a model that excluded that customer from training. The fold AP results matched the selected tuning configuration: 0.6581, 0.6946, 0.6879, 0.6361 and 0.6544.

The five validation-fold score sets were concatenated into one development table containing one OOF score per customer. Candidate thresholds were evaluated using F1 across this pooled table; the selected F1 was not an average of five separately optimised fold F1 values.

10.1 Development operating rule

The threshold maximising F1 on the selected configuration's OOF scores was 0.319015704046037. A customer is flagged when score ≥ threshold. An exact tie in maximum F1 would select the highest tied threshold. This is a development-defined rule; the final test was not used to choose it.

OOF scores were used instead of predictions from a model fitted on all the same development records because in-sample scores could favour an overly optimistic threshold. Selecting the highest threshold in an exact maximum-F1 tie makes the rule deterministic and, for fixed scores, avoids flagging more customers than a lower equally scoring threshold. It does not impose a stakeholder-approved capacity limit.

Table 14: Development OOF threshold results

OOF selection result	Value
Precision / recall / F1	0.5547 / 0.7525 / 0.6387
Flagged	2,028 of 5,634 (36.00%)
True positives / false positives	1,125 / 903
False negatives / true negatives	370 / 3,236

OOF predictions avoid training a model on the exact record it scores. However, configuration selection already used the same development folds. Choosing the threshold on these scores adds another selection step. We therefore describe the OOF metrics as selection evidence, not an unbiased independent evaluation or nested cross-validation result.

We used development cross-validation for selection and a separate reserved test set for final assessment. Nested cross-validation would provide an additional assessment of the selection process but would repeat the search within outer folds. It was outside the chosen evaluation design. Consequently, the selected development AP and OOF F1 are treated as selection results, while the reserved test provides the final assessment of the frozen procedure.

10.2 Freeze then fit then hand over

The group saved the complete estimator settings, predictor schema, positive class, threshold, flag rule, planned metrics and development evidence fingerprints. A fresh selected pipeline was then fitted on all 5,634 development records, producing 46 encoded features. The threshold remained fixed.

The fitted pipeline was saved as selected_churn_pipeline.joblib. Reload verification used 28 development records, including the eight with missing TotalCharges. Original and reloaded scores matched within numerical tolerance and high-risk flags matched exactly. The handover record identifies the model, preprocessing code and frozen procedure used by the leader.

The completed task used scores for ranking and a development-selected flagging rule, rather than for estimating expected financial losses from calibrated probabilities. Probability calibration was therefore outside the evaluated procedure. This does not establish that the probabilities are accurate, and calibration could affect the scores and threshold. Outputs are described as model scores; the threshold of approximately 0.319 is not an established real-world 31.9% future-churn risk boundary.

Evidence: selected_model_oof_scores.csv; selected_threshold_development_summary.csv; frozen_model_procedure.json; model_reload_verification.csv.

11. Reserved Test Evaluation

11.1 Checks before scoring

The leader verified the model, frozen procedure, shared module and development fingerprints against the handover records, and confirmed the recorded software versions. The original source fingerprint, saved split membership and source values were checked. No customer appeared in both subsets. TotalCharges blanks were normalised consistently for the source-value comparison.

The saved fitted pipeline was applied to all 1,409 test records without fit or fit_transform. The score for the positive class was extracted, and the frozen threshold was applied. The model and threshold were not revised after viewing test results. Re-executing the evaluation notebook reproduces this frozen calculation; it is not a new independent test sample.

Table 15: Reserved test results

Final measure	Result	95% bootstrap CI	Interpretation
Average precision	0.6565	0.6070–0.7087	Positive-class ranking on reserved records.
Precision	0.5491	0.5063–0.5927	285 of 519 flagged customers had recorded churn.
Recall	0.7620	0.7191–0.8042	285 of 374 recorded churners were identified.
F1	0.6383	0.6014–0.6748	Balance of precision and recall at the frozen threshold.
Flagged workload	519 / 1,409 = 36.83%	Not estimated	A substantial shortlist; capacity has not been verified.
False alerts	234	Not estimated	Recorded non-churners flagged for possible attention.
Missed churners	89	Not estimated	Recorded churners not flagged by this rule.
Correct non-churn classifications	801	Not estimated	Recorded non-churners correctly left unflagged.

Intervals are approximate 95% percentile bootstrap confidence intervals from 10,000 resamples of test customer records, with the model and threshold fixed. See Section 11.2.

The test AP is above the 0.2654 prevalence reference for a constant-score classifier on this test set. This supports useful discrimination within the supplied data. Test F1 is close to the development OOF F1, but closeness alone does not establish robustness across time, providers or populations.

Final test evaluation assessed the procedure selected using development data. The model and threshold were not revised using test outcomes. Logistic regression was not evaluated on the test set, so no claim is made that Random Forest outperformed it there. A supplementary analysis subsequently estimated confidence intervals for the frozen procedure, as described in Section 11.2.

Evidence: final_test_summary.csv; final_test_predictions.csv; final_evaluation_run_record.json.

11.2 Uncertainty of the frozen procedure

After the original final evaluation, we added an uncertainty analysis using the saved predictions for 1,409 reserved-test customers. We calculated approximate 95% percentile bootstrap confidence intervals for AP, precision, recall and F1 using 10,000 resamples and random seed 42. Each resample drew complete customer rows with replacement, preserving the association between the recorded label, score and frozen-threshold flag. Neither the model nor the threshold changed.

Table 15 reports AP of 0.6565 with an approximate 95% confidence interval of 0.6070–0.7087. The corresponding intervals are 0.5063–0.5927 for precision, 0.7191–0.8042 for recall and 0.6014–0.6748 for F1. These intervals describe sampling uncertainty around the reported performance. They do not describe uncertainty in an individual customer's score or whether contacting that customer would prevent churn.

The analysis assumes independent customer records and estimates test-sample uncertainty conditional on the fitted procedure. It excludes variability from retraining, model selection and threshold selection, and does not establish future-period performance or population representativeness. No formal between-model significance test was performed. The intervals therefore do not establish that Random Forest outperforms logistic regression; the small development AP difference remains descriptive selection evidence.

Evidence: 05_frozen_model_uncertainty.ipynb and the confidence-interval table, bootstrap metric values and run record under outputs/tables/final_evaluation/uncertainty_analysis/.

11.3 Final evaluation figures

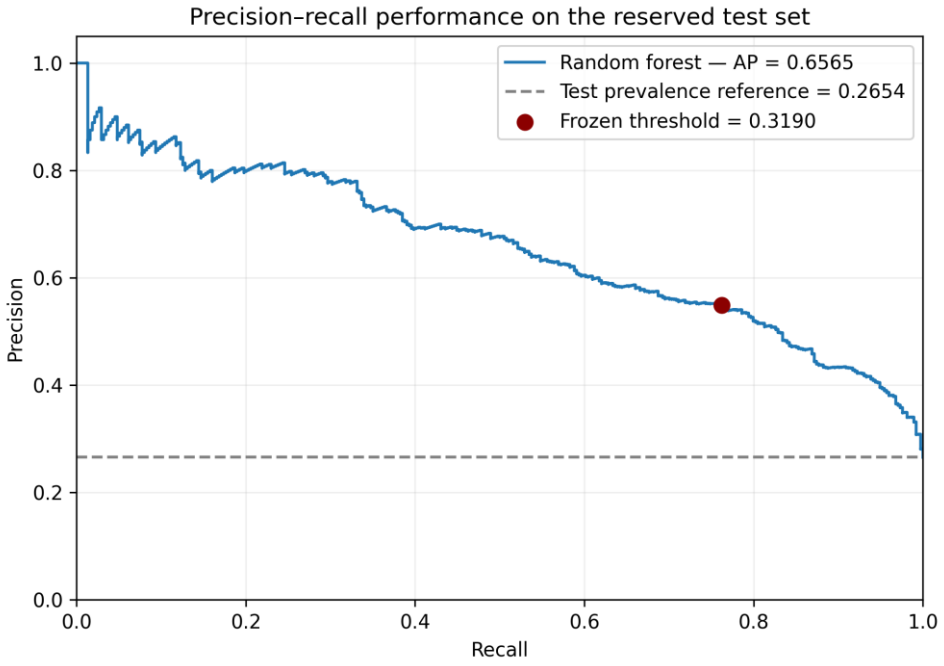


Figure 4: Test precision–recall curve. The red marker is the development-frozen operating threshold.

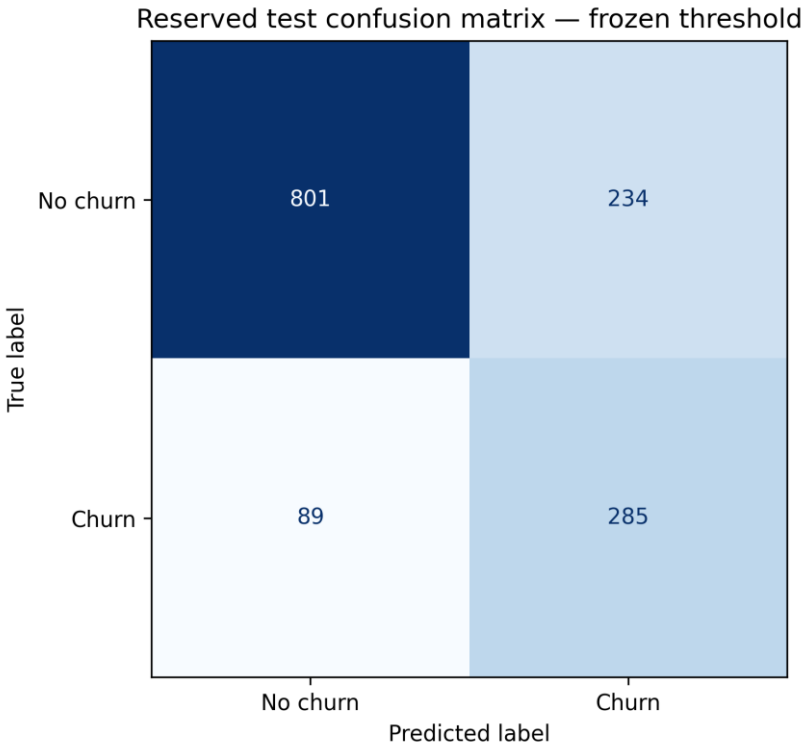


Figure 5: Test confusion matrix. Rows are recorded labels; columns are frozen-threshold predictions.

The curve displays the ranking trade-off; it was not used to choose a new threshold. The confusion matrix makes the two operational error types visible: 234 false alerts and 89 missed recorded churners. Neither error count measures whether an outreach intervention would succeed.

12. Descriptive high risk profiling

The leader joined test predictions to original test records by customerID, checking one-to-one membership and consistency of labels and flags. The secondary analysis compares the model-flagged and unflagged groups. It does not refit the classifier or select a new threshold.

Table 16: Test risk group outcomes

Model group	Customers	Recorded churners	Non-churners	Churn percentage
Unflagged	890	89	801	10.00%
Flagged	519	285	234	54.91%

The flagged group contains a much higher recorded churn proportion than the unflagged group. This describes the held-out sample under this rule; it is not an estimate of intervention benefit. The 10% recorded churn among unflagged customers means “unflagged” must not be interpreted as “no risk”.

Table 17: Numerical profiles by risk group

Feature	Unflagged median	Flagged median	Observed n (unflagged / flagged)
tenure	46.50 months	8.00 months	890 / 519
MonthlyCharges	58.35	78.75	890 / 519
TotalCharges	1,885.15	556.35	887 / 519

The numerical profile shows shorter relationships and higher monthly charges among flagged customers. Lower total charges are consistent with their shorter tenure, but this is a descriptive relationship rather than a verified billing mechanism or causal explanation.

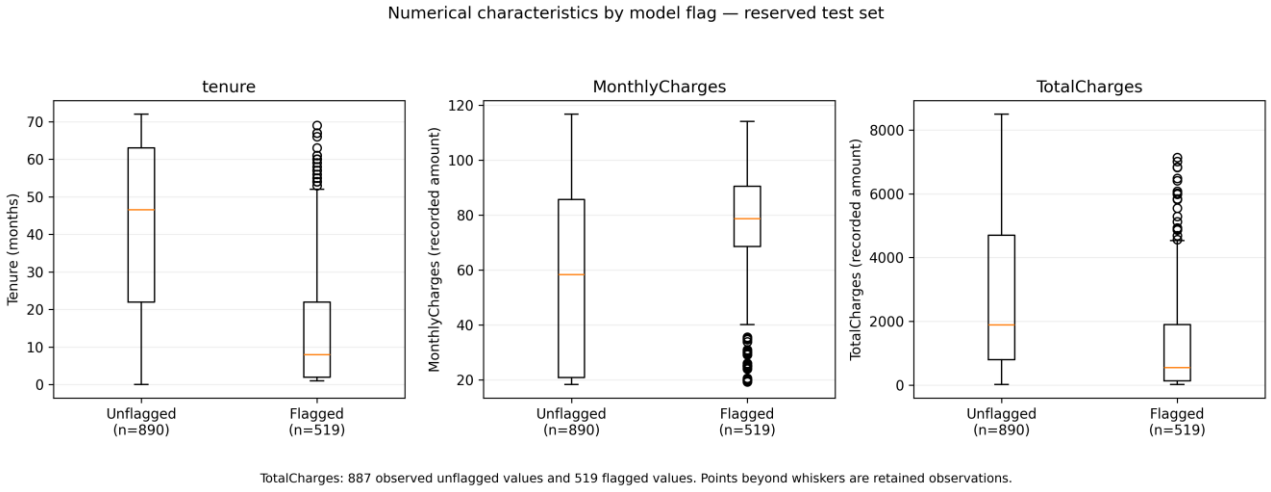


Figure 6: Numerical profiles by model flag. Charges are recorded amounts; points beyond whiskers are retained observations.

Three original test TotalCharges values are missing, all in the unflagged group. Prediction handled them through the development-fitted imputer. Profiling deliberately reports observed amounts: 887 available unflagged values and 519 flagged values. This avoids presenting imputed amounts as real customer charges.

12.1 Categorical profiles and their meaning

Table 18: Categorical profiles by risk group

Characteristic	Unflagged group	Flagged group
Month-to-month contract	29.33%	98.65%
One-year contract	32.92%	1.35%
Two-year contract	37.75%	0.00%
Fiber-optic internet	25.51%	74.37%
Electronic-check payment	18.09%	60.31%
No online security	28.65%	85.93%
No technical support	29.44%	84.78%

Each percentage uses the corresponding risk-group size as its denominator: 890 unflagged or 519 flagged. For example, 98.65% means that almost all flagged customers have month-to-month contracts. It does not mean that 98.65% of all month-to-month customers churned, or that every customer with that contract will be flagged.

Contract, InternetService, PaymentMethod, OnlineSecurity and TechSupport were useful descriptive characteristics identified through development EDA. Their profile contrasts are not a ranked feature-importance analysis. These characteristics were also inputs to the classifier, so differences between flagged and unflagged groups describe the model-generated shortlist rather than provide independent evidence about causes of churn. Profiling answers “What does the flagged group look like?”; it does not establish why a particular customer received a particular score. The same customer can appear in several categories across different variables; percentages across variables must not be added or treated as independent segments.

Using within-group percentages makes the 890-person and 519-person groups comparable despite unequal sizes; raw category counts alone would confound composition with group size. Medians and quartiles summarise the numerical distributions without being dominated by the largest recorded amounts. Keeping original missingness in the descriptive tables distinguishes measured customer information from the replacements needed for model computation.

No two-year-contract customer was flagged in this test subset. That does not establish that such customers never churn or should never receive support. Likewise, absence of online security or technical support does not prove that selling those services prevents churn. These features may overlap with contract, tenure, pricing and other observed or unobserved differences.

12.2 Implication for a retention discussion

The shortlist suggests questions about early customer experience, month-to-month relationships, charge expectations and service needs. A stakeholder could use these questions to design research or a monitored pilot. The project does not select an offer, establish affordability, calculate customer lifetime value or show which individual explanation caused a particular flag.

Evidence: test_profile_group_summary.csv; test_profile_numerical.csv; test_profile_categorical.csv.

Figure 7 below visualises the selected categorical profiles.

12.3 Categorical profile figure

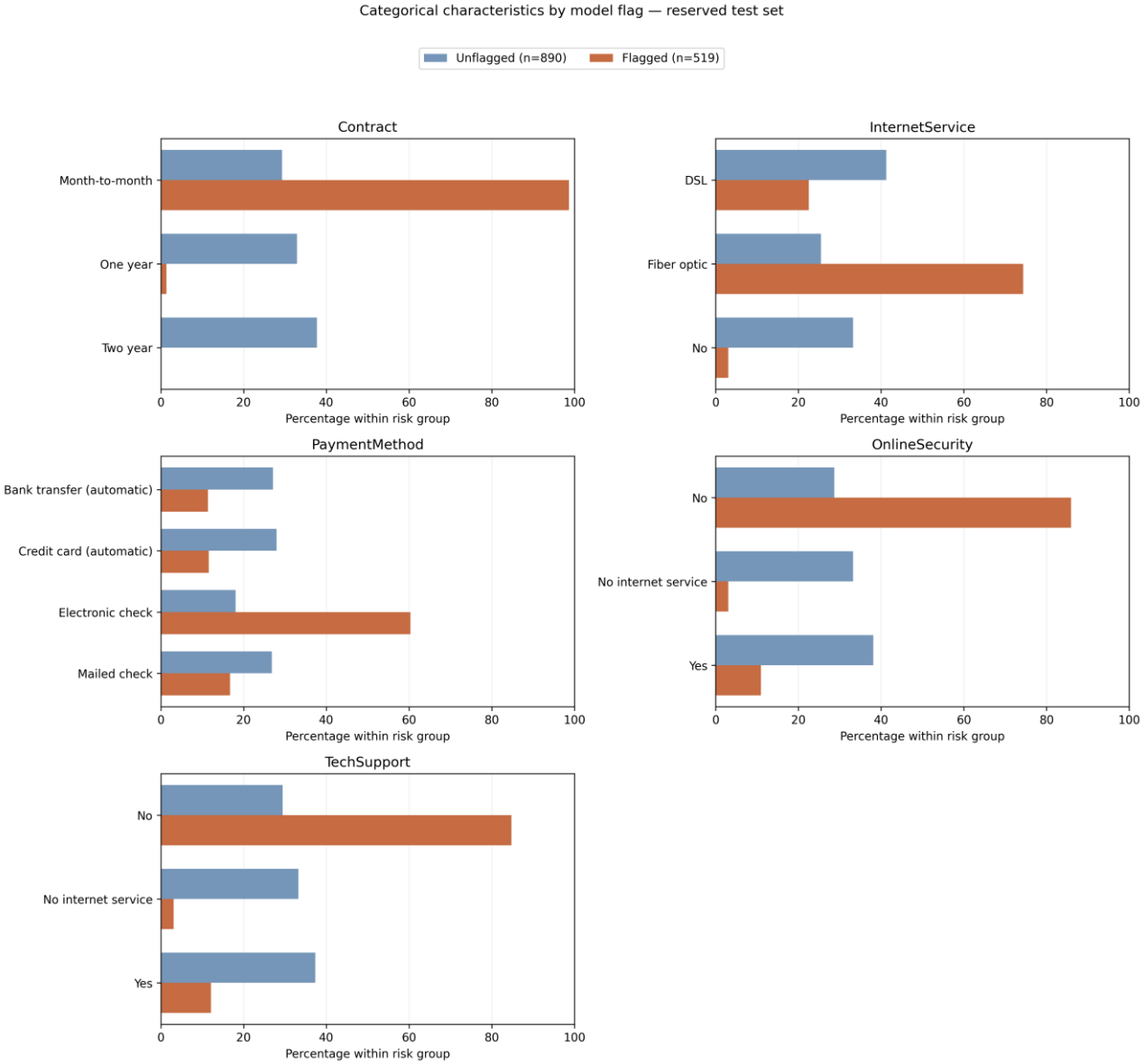


Figure 7: Category percentages within the flagged and unflagged test groups.

"No internet service" is kept separate from "No" in the add-on fields. These distinctions preserve service meaning. All charts describe the frozen model's output on the reserved test subset and are not used to change its decision rule.

13. Critical judgement and responsible use

Table 19: Limitations and responses

Limitation	Consequence	Response and remaining work
Unknown predictor timing and sampling	Random holdout is not prospective validation; prevalence may not represent a provider.	Disclose construction limits; obtain dated pre-outcome records before forecasting claims.
Prior whole-file descriptive exposure	The test was not wholly unseen in every descriptive respect.	Disclosed in split record; subsequent detailed development and tuning kept separate.
Bounded search and reused folds	Selected development AP can be optimistic; untested alternatives may perform better.	Use frozen test evidence; no global-optimality or significance claims.
Imputation and feature choices not ablated	Technically correct preparation is not proof of best treatment.	Document rationale and limitations; future experiments need a new development protocol.
Scores not calibrated	Numeric scores are not verified operational probabilities.	Use cautious score language; assess calibration on appropriate future validation data.
No fairness audit	Retained demographic fields and proxies may produce unequal errors or impacts.	Review subgroup performance, data use and stakeholder consequences before adoption.
Descriptive profiles only	Associations do not establish causes or effects of an offer.	Use profiles to form questions; evaluate interventions separately.
Unknown costs and capacity	F1 threshold may create excessive workload or poor economics.	Agree contact constraints and error costs before an operational policy.

Gender and SeniorCitizen were retained under the initial all-predictor comparison policy, not because their necessity or acceptable impact had been established. No subgroup fairness audit was completed. Before operational adoption, the provider should assess whether these attributes are needed, how errors vary across groups and whether outreach could produce unequal or exclusionary effects.

Identifiers were excluded from predictors and retained only for joins and traceability. The submission includes customer-level artefacts derived from the supplied public dataset for reproducibility. Any future provider data would require appropriate access controls and data-handling arrangements. A public demonstration should favour aggregate evidence rather than displaying individual records. No operational customer contact or automated adverse action was performed.

The secondary lens was implemented as descriptive profiling because its purpose was to characterise the shortlist produced by the primary model. SHAP and permutation importance would address a different question about model behaviour and were not implemented. Consequently, the report provides neither individual prediction explanations nor ranked feature attribution. It also makes no claim to estimate intervention effects, revenue or customer lifetime value, for which the required evidence was unavailable.

14. Recommendation and Conclusion

14.1 Recommendation supported by this project

Use the frozen random forest as a reproducible proof of concept for identifying records associated with recorded churn. It found 76.20% of recorded churners while flagging 36.83% of the reserved sample. Its precision of 54.91% means almost half of flagged records were recorded non-churners. A retention team must consider that workload and error trade-off before using the shortlist.

The descriptive profiles suggest that early-tenure, month-to-month relationships deserve investigation, alongside price and service experience. They do not justify automatically changing customers' contracts, pressuring customers to buy add-ons, or treating all flagged customers as equally preventable losses. Retention support should remain proportionate and subject to human judgement.

14.2 Conditions for moving beyond the proof of concept

Table 20: Conditions for operational validation

Next step	Purpose and evidence needed
Confirm the operational question	Agree an outcome horizon, eligible customer population, scoring date, contact capacity and acceptable error costs.
Obtain suitable dated data	Obtain timestamped customer histories, verify when each input became available, define observation and future outcome windows, and allow any required action delay. Check that customers have sufficient follow-up to assign outcome labels.
Validate on later records	Rebuild and train the procedure under the defined time-window design, then assess it on a later period. Review uncertainty, calibration and subgroup errors before operational use.
Set an operational policy	Compare workload and costs on appropriate development data; do not retroactively optimise this reported test result.
Evaluate a retention intervention	Use a controlled pilot to measure incremental retention, customer experience, cost and possible harms.
Monitor and govern	Track drift, unknown categories, missing fields, performance and fairness; record authorised revisions and retraining.

14.3 Final conclusion

Our primary decision lens produced a measurable classification procedure, and our secondary lens explained the composition of its shortlist. A fixed split, fold-specific preprocessing, a bounded model search, a frozen threshold and saved evidence made the work traceable across four members. The random forest was selected under the agreed AP rule, while the close logistic-regression result showed that additional complexity brought only a small development advantage.

The result supports further investigation of churn decision support. It does not establish future-period performance, a statistically superior algorithm, financially optimal prioritisation or proven retention gains. The final recommendation is therefore a controlled next-stage validation rather than immediate deployment.

15. Reproducibility and AI use

15.1 Execution environment and preserved evidence

Table 21: Recorded software environment

Component	Recorded version
Python	3.13.15
pandas	2.2.3
NumPy	2.1.3
scikit-learn	1.6.1
joblib	1.6.0

Table 21 records the original modelling and final-evaluation environment. The later Notebook 01 documentation rerun used Python 3.13.16, with pandas 2.2.3 and scikit-learn 1.6.1. That rerun reused the saved split and regenerated the dictionary and EDA outputs; it did not retrain or change the selected model.

The project runs in Google Colab with a shared Drive directory. The notebooks preserve source verification, split records, CV membership, settings, scores, figures and model files. File hashes connect handovers to the artefacts actually used. They detect content changes but do not independently establish that a method is valid.

15.2 Reproduction order

1. Mount Drive and point the project path to the shared root. Keep the raw CSV and saved split records unchanged. Run notebook 01 to verify the source and reuse existing membership, then reproduce development EDA and dictionary outputs.

2. Run notebook 02 to create or verify the shared preprocessing module and reproduce diagnostic exports. Do not hand over fitted diagnostic transformers for model comparison.

3. Run notebook 03 using the existing CV manifest and protocol. It repeats comparison, tuning, selection, OOF thresholding and the final development fit. Existing consistency checks should be investigated if they fail rather than bypassed by deleting records. Fit timings may vary.

4. Run notebook 04 with the saved pipeline and frozen procedure. It checks integrity, scores without fitting and reproduces final metrics, profiles and figures. The repeated run must preserve the selected procedure; it does not create fresh independent evidence.

5. Run Notebook 05 using the saved final test predictions and summary. Verify the reproduced point estimates, calculate the bootstrap intervals and save the analysis record. This step does not refit the model or reselect the threshold.

Core notebooks 01–04 passed fresh-session execution. The supplementary Notebook 05 also passed a fresh-session, top-to-bottom run on 11 October 2026. The submitted executed notebooks contain no stored error outputs. Run records reflect recorded execution times rather than an immutable audit trail of first access. Model files should be loaded only from the trusted project bundle.

15.3 AI use declaration

We used ChatGPT and Claude to help clarify concepts, discuss methodological alternatives, draft and debug code, interpret displayed outputs, and prepare documentation, including this report. AI assistance was iterative: suggested steps were run in Colab, outputs and assertions were inspected, and wording and scope were revised as the evidence developed.

AI-generated suggestions are not independent evidence. The saved data, executed notebooks, verification checks and exported results support the numerical claims. Group members remain responsible for the decisions, accuracy, source acknowledgement and their own understanding. This disclosure does not claim that any member independently authored every code line or that unperformed checks were completed.

16. Group contribution and evidence integration

Table 22: Member contributions and handovers

Member	Completed contribution	Key handover
Member 1 IT24103677	Coordinated framing and decision log; verified frozen-model handover; evaluated test records; produced profiles; estimated uncertainty from saved test predictions; integrated limitations and recommendation.	Notebook 04; Notebook 05 and uncertainty-analysis outputs; final evaluation and profile outputs; group review and final report.
Member 2 IT24102420	Verified source and data quality; saved and reused fixed split; analysed development data; reviewed dictionary definitions and logged insights.	Notebook 01; split and EDA evidence; reviewed dictionary.
Member 3 IT24103676	Chose and implemented parsing, imputation, encoding and model-specific scaling; verified training-only preparation and shared code behaviour.	Notebook 02; telco_preprocessing.py; PP01–PP06 and verification outputs.
Member 4 IT24102789	Compared baseline and three alternatives; ran bounded tuning; selected model and threshold; froze and fitted pipeline; verified reload.	Notebook 03; CV, search and OOF outputs; fitted pipeline and handover record.

16.1 How the work demonstrates the required capabilities

Problem framing connected a retention decision to a customer-level classification task. Critical thinking led us to remove the unsupported value proxy and distinguish recorded churn from future forecasting. Statistical reasoning shaped the split, positive-class metrics and cautious interpretation of small differences. Implementation joined reusable preparation to complete pipelines instead of treating cleaning and modelling as unrelated steps.

Communication and collaboration were supported by shared logs, explicit handovers and a reviewed recommendation. Learning how to learn involved tracing unfamiliar metrics and data definitions to documentation, checking AI suggestions against outputs and correcting earlier assumptions. Individual learning reports should explain each member’s own experience rather than duplicate this group narrative.

16.2 Assignment evidence coverage

The introduction and framing canvas are in Sections 1–2; scope and workflow are in Sections 3–4; data quality and EDA are in Sections 5–6; preprocessing reasoning and verification are in Section 7; model comparison and selection are in Sections 8–10 and Appendices A-C; final evaluation and profiling are in Sections 11–12; limitations and recommendation are in Sections 13–14. Section 15 covers reproducibility and AI use, and Section 16 covers contributions. The detailed logs and executable artefacts accompany the report.

The separate three-minute YouTube demonstration and each member's one-A4 Personal Learning Journey are submission components specified by the descriptor [1]. The submitted demonstration presents the completed workflow, saved pipeline, frozen evaluation and descriptive profiles.

16.3 Submission links

The supporting project files are available in the shared Google Drive folder below. It contains the final notebooks and shared preprocessing code, dataset and data dictionary, decision and technical logs, saved model and evaluation evidence. Appendix E identifies their locations. The model comparison, reasoning and conclusions are presented within this report.

Supporting project files: Open Group 2026-DS-77 project folder

Three-minute project demonstration: Watch the Group 2026-DS-77 video on YouTube.

The 2-minute 58-second demonstration presents the model comparison, saved pipeline, final evaluation and characteristics of the flagged customer group. The video is unlisted and accessible to anyone with the link.

References

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Primary project evidence

Group 2026-DS-77. Executed core notebooks 01–04 and supplementary uncertainty notebook 05, shared preprocessing module, split and model records, exported development and final evaluation tables, EDA Insight Log, Preprocessing and Feature Log, and Group Decision Log. These are the primary sources for this report's own experimental results. Appendix E maps the files to their purpose.

External studies supply context rather than comparable benchmark results. No external study's accuracy is used to validate this project's AP or F1. Source metadata and observed data values are kept distinct in the dictionary.

Appendix A Logistic regression search

Five shared development folds per configuration. Values are rounded for reporting; selection used unrounded mean validation AP. Configuration numbering follows the saved order. None means no class weighting.

Table 23: Logistic regression tuning results

Config	Parameters	Train AP	Valid AP	Fold SD
1	C=0.1; weight=None	0.6659	0.6596	0.0196
2	C=0.1; weight=balanced	0.6639	0.6584	0.0188
3	C=1.0; weight=None	0.6675	0.6607	0.0194
4	C=1.0; weight=balanced	0.6650	0.6594	0.0191
5	C=10.0; weight=None	0.6667	0.6601	0.0196
6	C=10.0; weight=balanced	0.6641	0.6585	0.0195

Configuration 3 was selected within this family. Changing regularisation and class weighting produced relatively small differences in mean AP in this bounded grid. The C = 1, unweighted setting matched the initial logistic comparator.

Evidence: tuned_model_cv_all_configurations.csv; full fold results in tuned_model_cv_fold_results.csv.

Appendix B Decision tree search

Five shared development folds per configuration. Values are rounded for reporting; selection used unrounded mean validation AP. Configuration numbering follows the saved order. None means no class weighting.

Table 24: Decision tree tuning results

Config	Parameters	Train AP	Valid AP	Fold SD
1	weight=None; depth=3; leaf=20	0.5732	0.5639	0.0225
2	weight=balanced; depth=3; leaf=20	0.5650	0.5538	0.0315
3	weight=None; depth=3; leaf=10	0.5732	0.5639	0.0225
4	weight=balanced; depth=3; leaf=10	0.5650	0.5538	0.0315
5	weight=None; depth=4; leaf=20	0.6146	0.6018	0.0250
6	weight=balanced; depth=4; leaf=20	0.6090	0.5941	0.0208
7	weight=None; depth=4; leaf=10	0.6146	0.6019	0.0249
8	weight=balanced; depth=4; leaf=10	0.6090	0.5941	0.0208
9	weight=None; depth=6; leaf=20	0.6836	0.6124	0.0213
10	weight=balanced; depth=6; leaf=20	0.6754	0.6087	0.0192
11	weight=None; depth=6; leaf=10	0.6850	0.6071	0.0245
12	weight=balanced; depth=6; leaf=10	0.6759	0.6060	0.0219

Configuration 9 was selected within this family. The deeper tree improved validation AP relative to the initial depth-4 tree, but the corresponding training–validation gap increased. Tied configurations at depth 3 illustrate why an explicit ordered tie rule was recorded.

Evidence: tuned_model_cv_all_configurations.csv; full fold results in tuned_model_cv_fold_results.csv.

Appendix C Random forest search

Five shared development folds per configuration. Values are rounded for reporting; selection used unrounded mean validation AP. Configuration numbering follows the saved order. None means no class weighting; an unrestricted depth is shown as None.

Table 25: Random forest tuning results

Config	Parameters	Train AP	Valid AP	Fold SD
1	weight=None; depth=10; leaf=10	0.7568	0.6662	0.0244
2	weight=balanced; depth=10; leaf=10	0.7539	0.6618	0.0245
3	weight=None; depth=10; leaf=5	0.7981	0.6627	0.0235
4	weight=balanced; depth=10; leaf=5	0.7903	0.6595	0.0261
5	weight=None; depth=None; leaf=10	0.7671	0.6643	0.0234
6	weight=balanced; depth=None; leaf=10	0.7665	0.6619	0.0249
7	weight=None; depth=None; leaf=5	0.8343	0.6592	0.0268
8	weight=balanced; depth=None; leaf=5	0.8333	0.6573	0.0266

Configuration 1 was selected both within the forest family and across the compared families. Limiting depth to 10 and requiring at least 10 training records per leaf improved mean validation AP over the unrestricted-depth, leaf-5 initial forest. The advantage over logistic regression remained small.

Evidence: tuned_model_cv_all_configurations.csv; full fold results in tuned_model_cv_fold_results.csv.

Appendix D Main decision trace

This index summarises the final decisions and links them to the detailed shared log. Historical reasons are preserved in that log. A completed technical check is distinct from evidence that a choice is optimal.

Table 26: Decision trace

Decision	Final position	Reason or boundary
D01	Guided Data Track	Supplied scenario and data; Industry Explorer approval and engagement not established.
D02–D03	Recorded-churn classification with descriptive profiling.	Direct binary target; secondary lens describes the resulting shortlist.
D04 / D04-R1	Value proxy removed	Past-spend approximation does not validate future value or intervention benefit.
D05–D06	No separate segmentation; customer-level binary task	Keep the analysis coherent and evaluate the available target.
D07–D08	Fixed 80/20 split; development-only detailed EDA	Consistent holdout and explicit disclosure of earlier whole-file exposure.
D09	Five shared stratified folds	Compare complete pipelines with preparation learned within training folds.
D10	Highest mean validation AP	Positive-class ranking; no claim of financial optimality.
D11	OOF F1 threshold; freeze then test	Separate selection from final scoring; no post-test retuning.
D12 / PP01	Training-only median imputation	Retain affected records; acknowledge tenure-zero realism limitation.
D13 / PP02–PP03	One-hot categories; model-specific scaling	Preserve category meaning and suit the classifier.
D14 / PP04–PP06	Retain all predictors and plausible observations; no resampling	No independent evidence for deletion or transformation; weighting compared in tuning.
D15	Dummy and three classifier alternatives	Compare no-discrimination, additive, single-tree and ensemble approaches.
D16	Add bootstrap confidence intervals for the frozen procedure	Quantify test-sample uncertainty without changing the model or threshold; does not establish superiority over logistic regression.

The group completed review of the original findings, limitations and stakeholder recommendation. The supplementary uncertainty analysis did not alter the model, threshold or predictions. The group reviewed the revised report, dictionary and supplementary analysis on 11/10/2026. Any later change to the selected procedure must be documented separately and evaluated appropriately.

Appendix E Evidence and submission package

Paths below are relative to the shared project root, ML 2026-DS-77 Telco Customer Churn. They identify the artefacts accompanying this report, rather than embedding thousands of individual predictions in the document.

Table 27: Project evidence inventory

Location	Contents and purpose
data/raw	Original WA_Fn-UseC_-Telco-Customer-Churn.csv, unchanged.
data/splits	development.csv, test.csv, split_manifest.csv and split_record.json; development_cv_manifest.csv and development_cv_settings.json.
data	dictionary_source_review.csv: reviewed meanings and unresolved details.
notebooks	01_data_verification_split_eda.ipynb; 02_preprocessing_feature_engineering.ipynb; 03_model_development_comparison.ipynb.ipynb; 04_final_evaluation_and_profiling.ipynb; 05_frozen_model_uncertainty.ipynb.
src	telco_preprocessing.py: shared fixed parsing and unfitted preparation factory.
models	selected_churn_pipeline.joblib; frozen_model_procedure.json; model_handover_record.json.
logs	Group Decision Log; EDA Insight Log; Preprocessing and Feature Log.
outputs/tables	Dictionary, EDA summaries, preprocessing checks, initial/tuned comparison, model selection, OOF scores, threshold comparison, feature names and reload check.
outputs/tables/ final_evaluation	Final test summary, predictions, confusion matrix, PR-curve data, evaluation record, three profiling tables and output inventory.
outputs/tables/ final_evaluation/ uncertainty_analysis	Confidence-interval table, bootstrap metric values and run record for the supplementary analysis of the frozen procedure.
outputs/figures	Development figures and final_evaluation figures for the PR curve, confusion matrix and numerical/categorical profiles.
report and references	Initial submission, workflow and supporting reading, with this final report as the integrated account.

The modelling notebook filename in the exported bundle contains a doubled .ipynb extension; the content is the executed modelling notebook. The final evaluation inventory records the expected result files. Its existence supports completeness checks, while substantive assertions and comparisons support correctness.

Full-size versions of the report figures are available in outputs/figures/ and its final_evaluation subfolder within the linked project folder.